

TATA, Hungary

WMO: 128360

Lat: 47.650N

Lon: 18.317E

Elev: 423

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.2	(c) 16.3	(d) 5.5	(e) 7.4	(f) 14.2	(g) 9.8	(h) 9.2	(i) 19.6	(j) 22.8	(k) 41.1	(l) 19.9	(m) 37.9	(n) 2.5	(o) 130	(p) 0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	92.3	71.0	89.2	69.7	86.1	68.5	72.8	87.5	71.4	85.5	69.9	83.3	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	105.1	78.8	66.6	99.5	77.0	65.2	94.8	75.5	36.6	87.7	35.3	85.6	34.1	83.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.6	(b) 16.0	(c) 13.9	(e) 5.4	(f) 97.2	(g) 6.4	(h) 3.0	(i) 0.8	(j) 99.3	(k) -3.0	(l) 101.1	(m) -6.5	(n) 102.8	(o) -11.2	(p) 104.9		
(a) 18.6	(b) 16.0	(c) 13.9	(e) 5.1	(f) 75.6	(g) 6.4	(h) 1.6	(i) 0.5	(j) 76.7	(k) -3.2	(l) 77.6	(m) -6.8	(n) 78.5	(o) -11.4	(p) 79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	31.7	34.9	43.2	53.8	61.4	68.5	71.6	70.3	61.5	52.3	43.6	34.2		
	DBStd	15.73	8.30	8.39	7.85	7.09	6.71	6.41	6.33	5.98	6.46	7.22	7.73	7.76		
	HDD50	2040	568	423	232	41	4	0	0	0	1	61	218	492		
	HDD65	5248	1033	842	676	341	149	37	15	18	144	396	641	956		
	CDD50	2893	0	1	22	154	358	555	668	630	345	132	27	1		
	CDD65	627	0	0	0	5	39	142	218	183	38	2	0	0		
	CDH74	6765	0	0	0	81	481	1494	2359	1935	398	17	0	0		
	CDH80	2578	0	0	0	12	116	523	1032	800	95	0	0	0		
	WSAvg	5.5	5.6	6.0	6.6	5.9	5.7	5.2	5.2	4.6	4.7	5.2	6.0	5.5		
	PrecAvg	23.0	1.4	1.3	1.3	1.5	2.5	2.4	2.7	2.3	2.2	1.8	2.0	1.6		
(15) Precipitation	PrecMax	38.8	3.0	4.6	4.3	3.4	8.1	5.5	6.5	4.8	5.6	4.3	4.1	4.7		
	PrecMin	14.2	0.1	0.0	0.0	0.0	0.6	0.2	0.5	0.0	0.3	0.1	0.2	0.2		
	PrecStd	5.2	0.7	1.0	0.9	0.8	1.7	1.3	1.5	1.3	1.4	1.1	0.9	1.1		
	PrecStd	5.2	0.7	1.0	0.9	0.8	1.7	1.3	1.5	1.3	1.4	1.1	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	59.2	69.8	81.6	87.7	92.6	97.7	96.9	87.4	76.2	65.7	56.0		
		MCWB	47.7	48.4	54.5	62.1	65.8	71.6	71.6	71.2	67.5	62.8	56.8	50.0		
	2%	DB	49.5	53.7	65.2	76.0	83.1	89.4	93.1	91.8	82.7	71.9	61.4	51.5		
		MCWB	44.8	46.0	51.9	58.1	65.0	70.1	71.3	70.8	65.4	61.0	54.7	47.3		
	5%	DB	45.8	50.3	60.6	72.2	79.5	86.1	89.7	88.1	78.7	68.5	57.8	47.5		
		MCWB	42.0	45.1	49.6	56.6	63.5	68.8	69.9	69.4	64.4	58.9	51.9	44.1		
	10%	DB	43.0	47.0	56.6	68.5	75.5	82.7	86.1	84.4	74.7	64.8	54.5	44.7		
		MCWB	39.9	42.7	47.7	54.6	61.2	67.3	68.8	67.9	62.0	57.1	49.9	41.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	49.3	55.4	63.0	68.5	74.4	74.5	74.5	69.3	64.3	57.7	50.4		
		MCDB	53.3	57.1	66.4	79.4	83.3	89.2	90.7	89.3	83.5	74.4	64.2	55.0		
	2%	WB	45.2	47.1	52.9	59.8	66.3	71.8	72.9	72.5	67.3	61.9	55.0	47.5		
		MCDB	49.1	52.7	62.8	73.2	80.0	85.3	88.4	86.9	79.6	70.3	60.9	51.3		
	5%	WB	42.5	45.1	50.7	57.4	64.4	70.1	71.4	71.0	65.3	59.9	52.5	44.5		
		MCDB	45.3	49.8	59.3	69.7	76.3	82.9	85.9	85.1	76.0	66.9	57.1	47.3		
	10%	WB	40.1	43.1	48.2	55.5	62.6	68.4	69.9	69.3	63.3	57.8	50.4	42.1		
		MCDB	42.7	46.9	55.7	66.8	73.8	80.1	83.3	82.1	72.4	63.8	54.1	44.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	13.2	17.8	21.4	20.7	21.3	22.5	22.4	20.6	17.6	12.1	10.1		
		MCDBR	14.7	18.2	25.7	27.7	27.4	26.0	28.0	28.2	26.6	23.3	15.7	12.6		
		MCWBR	10.7	12.2	14.6	12.9	12.0	10.4	10.3	10.7	12.2	13.2	10.3	9.6		
		MCDBR	14.1	16.6	23.7	25.8	24.5	23.3	24.7	25.3	23.6	21.4	14.6	12.5		
	5% WB	MCWBR	10.6	11.6	13.9	12.5	11.2	10.1	9.9	10.4	11.4	12.6	9.9	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.352	0.389	0.434	0.427	0.428	0.439	0.439	0.414	0.383	0.358	0.327		
	taud		2.363	2.320	2.246	2.138	2.210	2.244	2.215	2.227	2.263	2.328	2.353	2.378		
	Ebn at Noon		234	253	260	259	266	266	261	255	249	238	219	219		
	Edh at Noon		23	30	37	45	43	43	43	41	36	29	23	20		
(44) All-Sky Solar Radiation	RadAvg		326	583	974	1437	1728	1876	1831	1640	1159	734	384	248		
	RadStd		39	85	142	140	183	142	139	159	139	79	39	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		+1.08	N/A	N/A	N/A	+0.97	+0.90	-223	-322	+182	+97

Nomenclature: See separate page